

K-FLARE: A knowledge-fused language and relational-learning framework for rumor detection

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ABSTRACT

Detecting social media rumors in their early stages, when user interaction is limited, remains a critical challenge. Most existing methods determine veracity by analyzing reply content, but that is ineffective when such signals are sparse or unavailable. To overcome this limitation, we propose K-FLARE, a novel dual-stream hybrid reasoning framework for reply-free rumor detection. K-FLARE's core theoretical contribution is a mechanism that synergizes a data-driven inductive stream, which learns from multimodal features (semantics, knowledge, topology, and temporality), with a knowledge-driven deductive stream, which leverages language models to directly assess content authenticity. This dual-stream approach effectively compensates for missing user replies. Extensive experiments on the Twitter15, Twitter16, and PHEME datasets demonstrate the superiority of K-FLARE. It achieves F1-scores of 92.01%, 92.58%, and 87.64% respectively, significantly outperforming state-of-the-art reply-free methods. Ablation studies confirm that each component is critical for compensating for the absence of reply information. Moreover, K-FLARE offers enhanced interpretability by identifying key rumor-indicative features, providing a robust solution for early rumor intervention.

1. Introduction

1.1. Research background and challenges

The proliferation of social media has enabled unprecedented speed in information dissemination, yet it has also become a fertile ground for rumor propagation. As unverified information, rumors can incite social panic, inflict economic losses, and undermine public health initiatives, as evidenced by the widespread misinformation during the COVID-19 pandemic[1]. Consequently, developing robust, automated rumor detection systems is critical for maintaining a healthy information ecosystem. However, research in this domain confronts three primary challenges that limit the practical efficacy of existing technologies: (1)The Early Detection Dilemma: Detecting rumors is most critical in their initial stages, precisely when direct evidence is scarcest. In this context, “early stage” refers to the initial phase of a post’s lifecycle (e.g., the first few hours) where user interaction, particularly the volume of textual replies, is minimal or non-existent. (2)Information Insufficiency: Many models depend on user replies for veracity assessment. In real-world, early-stage scenarios, these replies are often unavailable, insufficient, or of low quality. (3)Multimodal Fusion Barriers: Rumors are characterized by heterogeneous data, including text, propagation structures, and

temporal dynamics. Effectively integrating these disparate modalities remains a significant technical hurdle.

1.2. Existing methods and limitations

Prevailing paradigms in rumor detection can be broadly categorized into content-based and propagation-based approaches[2]. Content-based methods leverage NLP techniques to analyze textual features such as sentiment, topic, and linguistic style[3,4]. While effective when rich reply data is available, their performance degrades significantly in early-stage or low-interaction scenarios. Propagation-based methods model the diffusion of information, typically using Graph Neural Networks (GNNs) to analyze the structure of propagation trees[5,6]. These methods excel at capturing structural patterns but require a substantial volume of propagation data to be effective. Hybrid models that fuse both content and propagation features have shown improved accuracy[7,8], but their reliance on reply content persists, limiting their applicability.

Although recent studies have begun to incorporate knowledge graphs[9] and Large Language Models (LLMs)[10], a critical research gap remains: the lack of focus on effective detection in the absence of user replies. Addressing this gap is imperative for developing practical systems capable of early intervention against low-interaction rumors.

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1.3. Contributions of this study

To address the aforementioned limitations, this paper introduces K-FLARE, a novel “reply-free” rumor detection framework. Our work makes the following primary contributions:

1. **A Novel Dual-Stream Hybrid Reasoning Framework:** We propose K-FLARE, a novel framework designed for the reply-free scenario. Its core theoretical contribution is a dual-stream architecture that synergizes two complementary reasoning paradigms: (a) **A Data-Driven Inductive Stream:** This stream performs bottom-up reasoning. It learns emergent patterns by fusing a comprehensive set of multimodal features, including: deep semantics, explicit external knowledge, propagation topology, and temporal dynamics. This stream fuses these disparate data types to induce a pattern of veracity. (b) **A Knowledge-Driven Deductive Stream:** This stream performs top-down reasoning in parallel. It leverages the vast implicit world knowledge within LLMs and NLI models to deductively assess the authenticity of the source post content directly.
2. **An Evidence-Balancing Mechanism:** We introduce a weighted decision module that acts as an evidence-balancing mechanism. Its function is to integrate the outputs from the inductive data stream and the deductive knowledge stream to achieve a robust final prediction.
3. **Comprehensive Experimental Validation:** We conduct rigorous experiments on three benchmark datasets (Twitter15, Twitter16, and PHEME), demonstrating that our reply-free method consistently outperforms state-of-the-art methods when they are adapted to the same challenging, reply-free conditions.

This study advances the theoretical understanding of rumor detection and provides a practical, robust solution with significant application value for identifying rumors in their early, low-interaction stages.

2. Related work

2.1. Content-Based rumor detection

Content-based methods for rumor detection primarily analyze the semantic and stylistic features of text to ascertain its veracity. Early research in this domain relied heavily on handcrafted features. For instance, Castillo et al.[11] pioneered an approach that utilized a comprehensive set of 68 features spanning message content, user profiles, and propagation patterns to classify the credibility of Twitter posts.

With the advent of deep learning, the focus shifted towards automated feature extraction using neural networks. Ma et al.[12] introduced a model based on Recurrent Neural Networks (RNNs), which treated a source post and its subsequent replies as a time series. By employing Gated Recurrent Unit (GRU) networks, their method captured the temporal evolution of the conversation to identify rumor content. The emergence of Pre-trained Language Models (PLMs) marked another significant breakthrough. Kaliyar et al.[13] demonstrated that leveraging deep contextual representations from BERT could substantially improve detection accuracy. Building on this, Miao et al.[14] proposed syntax and sentiment enhanced BERT (SSE-BERT), which achieves superior performance in early rumor detection using only source posts by learning extra syntax and sentiment features.

A significant drawback of these methods, however, is their heavy reliance on the availability and quality of user replies. As a systematic study by Yu et al.[15] highlighted, their performance degrades sharply in the absence of sufficient reply content. Our research aims to overcome this limitation by proposing a detection framework that operates effectively without depending on user replies.

2.2. Propagation-Based rumor detection

Propagation-based approaches shift the focus from textual content to the diffusion patterns of information across social networks. These methods are founded on the principle that rumors and credible news exhibit distinct propagation dynamics. An early analysis by Wu and Liu[16] revealed that rumors tend to propagate faster and wider, especially during their initial stages. The development of Graph Neural Networks (GNNs) enabled a more sophisticated modeling of these dynamics. Ma et al.[17] were the first to represent the propagation process as a tree structure, using a tree-based RNN to capture the hierarchical dependencies between parent and child nodes in a reply chain. Subsequent research has explored more powerful GNN architectures. For example, the BiGCN model proposed by Bian et al.[8] captures information flow in both top-down (source to replies) and bottom-up (replies to source) directions. More recently, Choi et al.[18] introduced a dynamic GNN with an attention mechanism to model both the structural and temporal evolution of rumor propagation, thereby enhancing detection performance.

Although these methods are not directly dependent on textual content, they require sufficient propagation data to construct a meaningful graph structure. Furthermore, content and propagation features are inherently complementary; relying on either in isolation is unlikely to yield optimal results. Drawing on this insight, our work devises a multimodal fusion framework that synergistically integrates both content and propagation characteristics.

2.3. Knowledge-Enhanced rumor detection

To mitigate the limitations of methods that rely solely on text or propagation patterns, researchers have increasingly turned to external knowledge to enhance the accuracy and interpretability of rumor detection. By providing essential background context and enabling common-sense reasoning, external knowledge graphs (KGs) are particularly valuable in data-scarce scenarios. Haque et al.[19] pioneered the integration of the ConceptNet knowledge graph, demonstrating that constructing local knowledge subgraphs related to the post's content improves the model's ability to assess its veracity. Similarly, Xie et al.[20] proposed a contrastive learning framework that leverages complementary knowledge from Wikidata and DBpedia to augment a heterogeneous graph, achieving superior fake news detection performance by fusing the enhanced representations.

However, existing knowledge-enhanced methods have primarily focused on fusing external knowledge with existing text or propagation features. The potential of using KGs to specifically compensate for the information loss caused by missing reply content remains underexplored. As noted in the study by Li et al.[21], external knowledge could potentially serve as a proxy for user replies. However, while prior works have integrated KGs, their systematic application as a direct substitute for reply content within a multi-modal fusion architecture for reply-free scenarios remains a developing area. Our research is predicated on this observation, exploring how to effectively employ knowledge graphs as a substitute for reply content in resource-limited settings.

2.4. Application of large language models

The advent of Large Language Models (LLMs) has introduced a new paradigm for rumor detection. Unlike traditional models that depend on task-specific feature engineering, LLMs harness their vast world knowledge and sophisticated language understanding to evaluate content credibility directly. Foundational studies, such as the work by Wei et al.[22], have shown that models like GPT can achieve competitive performance in zero-shot and few-shot rumor detection scenarios, often rivaling models explicitly fine-tuned for the task. Subsequent research has focused on refining the innate capabilities of LLMs, with a key emphasis on interpretability. For instance, Nan et al.[23] demonstrated that

LLMs can be prompted to generate plausible, human-readable explanations for their classifications, addressing the “black-box” nature of many deep learning models and enhancing user trust.

Despite these advances, much of the literature treats LLMs as standalone classifiers, thereby overlooking the structural dynamics of information propagation. Recognizing that GNNs excel at modeling such structured information while LLMs provide deep semantic reasoning, a promising research frontier has emerged focused on integrating these complementary strengths. Our work is situated at this frontier, proposing a novel framework that deeply fuses the semantic power of LLMs with the structural modeling capabilities of GNNs to create a more holistic and effective rumor detection system.

2.5. Research gaps and positioning of this study

While integrating semantic models with propagation structures is a clear trend, a critical analysis of the latest advances in 2024–2025 reveals a persistent research gap that our work directly addresses. Recent research on this front has produced powerful rumor detection frameworks. Wang et al.[24], with LLM-GAN, and Wan et al.[23], with DELL, have successfully leveraged the generative and analytical power of LLMs. Others have focused on structural integration: Moorthy et al.[25] developed a dual stream transformer model, while Malik et al.[26] created ensemble GNNs to better model user engagement. A third paradigm, introduced by Mavromatis and Karypis[27] with GNN-RAG, uses GNNs as subgraph reasoners to augment LLM prompts. Despite their architectural diversity, these state-of-the-art methods share a unifying limitation: they are all designed for data-rich environments. Their effectiveness fundamentally relies on a substantial volume of replies, comments, and retweets to construct meaningful propagation graphs. This reliance severely limits their applicability in the earliest stages of a rumor’s lifecycle, precisely when intervention is most critical. K-FLARE is designed to fill this specific gap. It is a systematic, reply-free framework that operates without the user interaction signals that current LLM-GNN methods depend on. Our work thus offers a distinct and complementary approach, demonstrating that effective detection is possible even in the data-sparse conditions of an emerging rumor. As illustrated in Fig. 1, the lifecycle of a rumor can be divided into an early stage with sparse user interaction and a later stage with rich interaction. While state-of-the-art methods are effective in the data-rich later stage, they are not applicable during the critical early window. K-FLARE is strategically designed to operate within this early, data-sparse environment, addressing a crucial gap in existing research.

3. Methodology

3.1. Problem definition and framework overview

This section formally defines the rumor detection task and presents an overview of our proposed framework.

3.1.1. Problem definition

We formulate the rumor detection task as a binary classification problem. Given a source social media post m and its corresponding propagation tree $\mathcal{T}$, the objective is to learn a function f that classifies m as a rumor (label 0) or non-rumor (label 1). The propagation tree $\mathcal{T}$ is represented as a set of edges $\{e_1, e_2, \dots, e_n\}$, where each edge $e_i = (u_i^p, u_i^c, t_i^p, t_i^c)$ signifies that a post from user u_i^p at time t_i^p was propagated by user u_i^c at time t_i^c . Crucially, our formulation operates using only the source post content m and the propagation tree $\mathcal{T}$, without relying on any reply content. It is important to note that our “reply-free” formulation does not imply the propagation tree $\mathcal{T}$ is empty. It means that $\mathcal{T}$ may be sparse (typical of the early stage) and, crucially, that the model only uses the structure of the tree (i.e., propagation links) and the content of the source post m . No textual content from any reply nodes is used.

Formally, the task is defined in Eq. 1.

$$f(m, \mathcal{T}) \rightarrow y \in \{0, 1\} \quad (1)$$

where f is the detection model we aim to learn, with $y = 0$ indicating a rumor and $y = 1$ representing a non-rumor.

3.1.2. K-FLARE Framework overview

The K-FLARE framework is theoretically grounded as a dual-stream hybrid reasoning system. It comprises (1) a Data-Driven Inductive Stream, which includes the Multimodal Feature Extraction and Neural Fusion modules; (2) a Knowledge-Driven Deductive Stream, represented by the Hybrid Model for Language Reasoning Module. A final Weighted Decision Module serves as an evidence-balancing mechanism between these two streams. As shown in Fig. 2, K-FLARE consists of five core modules.

1. **Data Processing Module:** Extracts essential information from raw social media posts, including text cleaning and propagation tree construction.
2. **Multimodal Feature Extraction Module:** Integrates four parallel components to generate a comprehensive feature set:
 - (a) *Semantic Feature Extractor:* Employs BERT for in-depth semantic analysis of the source post content.
 - (b) *Commonsense Knowledge Extractor:* Constructs a knowledge sub-graph using ConceptNet to incorporate external world knowledge.
 - (c) *Topological Feature Extractor:* Analyzes the structural properties of the propagation network.
 - (d) *Temporal Feature Extractor:* Captures the temporal dynamics of information diffusion.
3. **Hybrid Model for Language Reasoning Module:** leverages hybrid models for advanced reasoning through two components:
 - (a) *LLM Content Analysis:* Performs nuanced semantic analysis via structured prompting, as shown in Fig. 3.
 - (b) *Zero-shot Classification:* As shown in Fig. 4, this module determines the veracity of the post directly.
4. **Neural Fusion Module:** Adopts a multi-level attention architecture to integrate the multimodal features and generate a unified prediction.
5. **Evidence-Balancing Mechanism:** Dynamically combines the predictions from the Neural Fusion Module and the Hybrid Model for Language Reasoning Module to produce the final classification result.

3.2. Multimodal feature extraction

To mitigate the information loss caused by the absence of reply content, we designed four complementary feature extractors to analyze the source post and its propagation patterns from diverse perspectives.

3.2.1. Semantic feature extraction

This module captures rich semantic information from the source post’s text using a pre-trained BERT model. The entire process of tokenization, contextual embedding, and feature extraction is analytically defined as the semantic feature h_{semantic} in Eq. 2, which utilizes the final hidden state of the $[CLS]$ token from the BERT output.

$$h_{\text{semantic}} = \mathbf{H}_{[CLS]} = \text{BERT}(\text{Tokenizer}(m))_{[CLS]} \in \mathbb{R}^{d_{\text{hidden}}} \quad (2)$$

where $\text{Tokenizer}(\cdot)$ performs the input text tokenization, $\text{BERT}(\cdot)$ computes the contextualized embeddings, and $\mathbf{H}_{[CLS]}$ is the representation vector corresponding to the $[CLS]$ token, which serves as the global semantic feature.

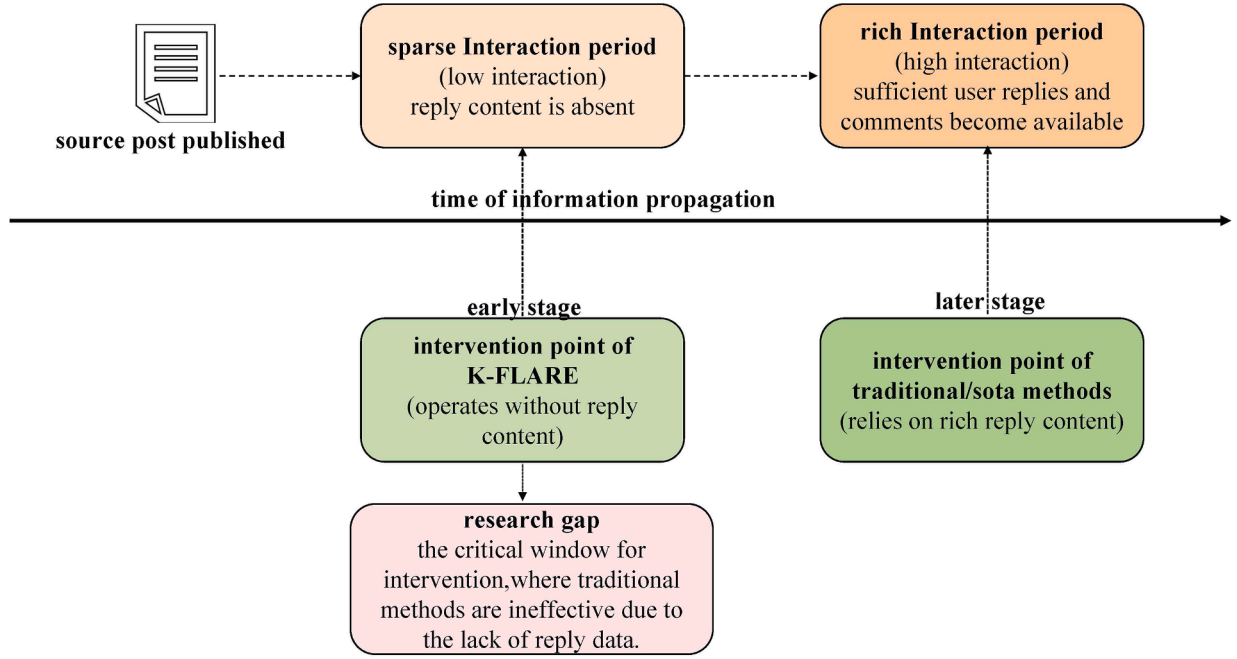


Fig. 1. A conceptual timeline of rumor propagation, illustrating the research gap addressed by K-FLARE. Traditional methods require the rich user interaction data found in the later stages, whereas K-FLARE is designed to intervene in the critical, data-sparse early stage.

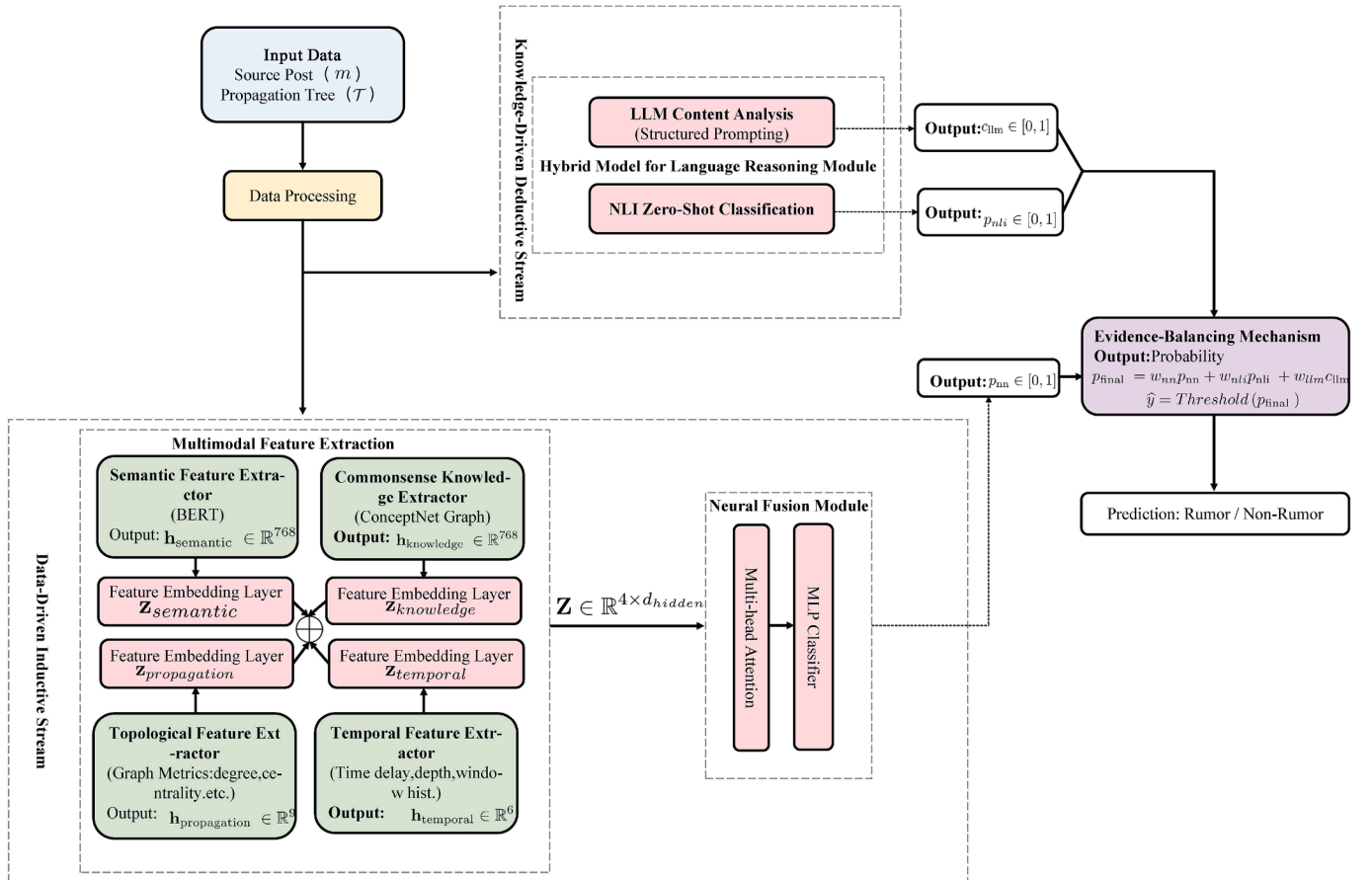


Fig. 2. K-FLARE framework architecture.

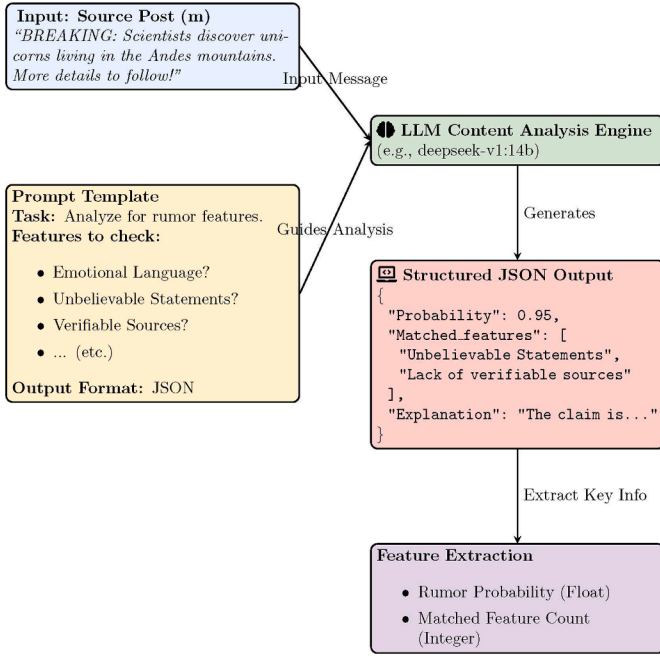


Fig. 3. LLM Content Analysis Module Workflow.

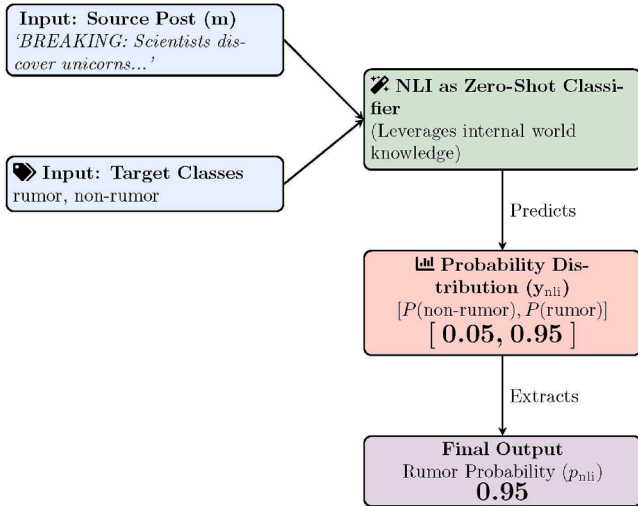


Fig. 4. Zero-shot Classification Module Workflow.

3.2.2. Commonsense knowledge-Enhanced feature extraction

To introduce external context and commonsense reasoning, we construct and leverage a knowledge graph. First, we utilize a pre-trained Named Entity Recognition (NER) model to identify key entities C from the source post m . These extracted entities are then used to query the ConceptNet API [28] to retrieve relevant semantic relations, thereby constructing a local knowledge subgraph $G_K = (V_K, E_K)$. To obtain a single graph-level representation for G_K , we generate a semantic embedding for each node $v \in V_K$ using BERT, and then aggregate these embeddings via mean pooling. This process is defined in Eq. 3.

$$h_{knowledge} = \frac{1}{|V_K|} \sum_{v \in V_K} \text{BERT}(v) \in \mathbb{R}^{d_{hidden}} \quad (3)$$

3.2.3. Topological feature extraction from propagation networks

This extractor analyzes the topological structure of the message propagation network to capture its structural characteristics. From the propagation tree $\mathcal{T}$, we construct a directed graph $G = (V, E)$, where V represents users and E represents information diffusion paths. We then ex-

tract a vector of graph-based features designed to create a comprehensive structural fingerprint of the propagation network. These features include:

1. **Scale Features** (Number of nodes ($n_{nodes} = |V|$) and edges ($n_{edges} = |E|$)): To quantify the overall size and volume of the propagation.
2. **Degree Distribution Features** (average/max in-degree and out-degree): To characterize user engagement patterns and the broadcast nature of the diffusion.
3. **Centrality Metrics** (average closeness, betweenness, and PageRank): To measure the influence and structural importance of nodes within the information flow.

This combination of features captures the network's macro-level scale, micro-level user interactions, and core structural dynamics as shown in Eq. 4.

$$\mathbf{h}_{propagation} = [n_{nodes}, n_{edges}, d_{avg}^{in}, d_{max}^{in}, d_{avg}^{out}, d_{max}^{out}, c_{avg}, b_{avg}, p_{avg}] \in \mathbb{R}^{d_{propagation}} \quad (4)$$

where $d_{propagation}$ is the dimension of the resulting feature vector.

3.2.4. Graph temporal feature extraction

This module captures the temporal dynamics of the propagation process. This is predicated on the hypothesis that rumors exhibit explosive, bursty propagation patterns, whereas non-rumors diffuse more uniformly over time. For each edge $e_i = (u_i^p, u_i^c, t_i^p, t_i^c)$ in the propagation tree, we compute the propagation delay time $\Delta t_i = t_i^c - t_i^p$. To quantify these patterns, we extract the following features:

1. **Basic Statistics**: The average (t_{avg}) and maximum (t_{max}) propagation delays: To capture the overall speed and potential bottlenecks in information diffusion.
2. **Propagation Depth**: The maximum depth (d_{max}) of the propagation tree: To measure the depth of the conversation and the length of the information cascade.
3. **Time-Window Distribution**: The propagation timeline is divided into K uniform time windows (where $K = 12$), and the activity count in each window is computed to form a normalized histogram h_{hist} : To model the temporal dynamics and identify characteristic propagation patterns, such as sudden bursts of activity common to rumors.

These features are concatenated to form the final temporal feature vector $\mathbf{h}_{temporal}$ as defined in Eq. 5.

$$\mathbf{h}_{temporal} = [t_{avg}, t_{max}, d_{max}, h_{hist}] \in \mathbb{R}^{d_{temporal}} \quad (5)$$

3.3. Hybrid model for language reasoning module

This module leverages the Hybrid Model Reasoning Module to perform a deep semantic analysis of the source post. We designed two complementary tasks: structured content analysis and zero-shot classification.

3.3.1. LLM Content analysis

We employ a structured prompt to guide the LLM in identifying common rumor-indicative features within the post content. The prompt instructs the model to assess features such as emotional language, unverifiable sources, and sensationalism, and to return a structured JSON object containing a Probability score, a list of Matched_features, and a brief Explanation. By invoking the LLM API, we obtain a structured analysis result, from which we extract the Probability score, and the number of matched features as inputs for the final decision stage.

3.3.2. Zero-shot classification

In parallel with the structured content analysis, we perform zero-shot classification by reframing the task as a Natural Language Inference (NLI) problem. This standard approach leverages a model pre-trained

on NLI datasets, such as MNLI, to determine the logical relationship between a “premise” and a “hypothesis”. For our task, the source post content m serves as the premise. We then craft a set of candidate labels as hypotheses, for instance, “This text is a rumor” and “This text is not a rumor”. The NLI model takes the premise-hypothesis pair $(m, hypothesis)$ as input and outputs a probability distribution over three NLI labels: entailment, neutral, and contradiction. The probability of entailment for the “rumor” hypothesis is treated as the rumor probability score p_{nli} . This method allows the model to assess the veracity of the post without being explicitly fine-tuned on rumor detection data. The process is formulated in Eq. 6.

$$p_{nli} = P(\text{entailment} \mid \text{premise} = m, \text{hypothesis} = \text{“This is a rumor”}) \quad (6)$$

3.4. Neural fusion module

This module uses a multi-level neural architecture to integrate the multimodal features and learn their complementary relationships.

3.4.1. Feature embedding layer

First, we project the four feature vectors h_i into a common high-dimensional space $\mathbb{R}^{d_{hidden}}$ using separate linear transformations. This process facilitates subsequent processing by the attention mechanism. The transformation for each feature vector h_i is defined by a dedicated linear layer, batch normalization (BN), and ReLU activation, as shown in Eq. 7.

$$Z_i = \text{ReLU}(\text{BN}(W_i h_i + b_i)) \in \mathbb{R}^{d_{hidden}} \quad (7)$$

where $h_i \in \{h_{semantic}, h_{knowledge}, h_{propagation}, h_{temporal}\}$, $W_i \in \mathbb{R}^{d_{hidden} \times d_i}$ and $b_i \in \mathbb{R}^{d_{hidden}}$ are the learnable weight matrix and bias vector for the linear transformation of feature i .

3.4.2. Multi-head attention mechanism

The embedded features are stacked into a matrix $\mathbf{Z} \in \mathbb{R}^{4 \times d_{hidden}}$, as defined in Eq. 8.

$$\mathbf{Z} = [z_{semantic}; z_{knowledge}; z_{propagation}; z_{temporal}] \quad (8)$$

We then apply a multi-head self-attention mechanism to dynamically learn the inter-feature relationships. The standard attention mechanism is defined in Eq. 9.

$$\begin{aligned} \mathbf{Q} &= \mathbf{Z}\mathbf{W}^Q, \mathbf{K} = \mathbf{Z}\mathbf{W}^K, \mathbf{V} = \mathbf{Z}\mathbf{W}^V \\ \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \end{aligned} \quad (9)$$

The multi-head variant performs this operation in parallel across multiple heads and concatenates the results, as shown in Eq. 10.

$$\text{MultiHead}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^O \quad (10)$$

where $\text{head}_i = \text{Attention}(\mathbf{Z}\mathbf{W}_i^Q, \mathbf{Z}\mathbf{W}_i^K, \mathbf{Z}\mathbf{W}_i^V)$. This allows the model to simultaneously attend to different feature interaction patterns, adaptively weighing each modality based on the sample’s characteristics.

3.4.3. Feature integration and output layer

The output from the multi-head attention mechanism Z_{attn} , is flattened into a fused representation, $Z_{fused} = \text{Flatten}(Z_{attn})$. This fused vector is then passed through a multi-layer perceptron (MLP) for final classification. The MLP consists of two linear layers with parameters (W_1, b_1) and (W_2, b_2) , yielding the neural network’s prediction y_{nn} as defined in Eq. 11.

$$y_{nn} = \text{softmax}(W_2(\text{ReLU}(\text{BN}(W_1 Z_{fused} + b_1))) + b_2) \in \mathbb{R}^2 \quad (11)$$

3.5. Evidence-Balancing mechanism

The final module serves as a principled evidence-balancing mechanism, designed to integrate the outputs from K-FLARE’s two distinct reasoning streams. It addresses the core challenge of how to best combine evidence from the data-driven inductive stream (which produces the probability p_{nn}) with evidence from the knowledge-driven deductive stream (which produces probabilities p_{nli} and c_{llm}). Rather than a simple heuristic, this module implements an optimized fusion strategy. The final rumor probability, p_{final} , is calculated via a weighted sum, which acts as a linear combination of the evidence from each stream as defined in Eq. 12.

$$p_{final} = w_{nn} \cdot p_{nn} + w_{nli} \cdot p_{nli} + w_{llm} \cdot c_{llm} \quad (12)$$

where the weights $(w_{nn}, w_{nli}, w_{llm})$ are not arbitrary but are empirically optimized hyperparameters. They are learned via 5-fold cross-validation on the training set with the objective of minimizing the final cross-entropy loss. This process finds the optimal balance for a given dataset, determining how much to trust the data-based inductive evidence versus the knowledge-based deductive evidence to achieve the most robust prediction. The final classification $\hat{y}$ is determined by applying a threshold of 0.5 to p_{final} , as shown in Eq. 13.

$$\hat{y} = \begin{cases} 0, & \text{if } p_{final} > 0.5 \\ 1, & \text{otherwise} \end{cases} \quad (13)$$

This approach ensures the weights are systematically determined via empirical optimization. This process is formally defined as an optimization problem in Appendix B.

3.6. Model training and optimization

We train the neural fusion module end-to-end while keeping the LLM parameters frozen. The training procedure involves adopting the cross-entropy loss function, as formulated in Eq. 14, which includes $L2$ regularization.

$$\begin{aligned} \mathcal{L}_{CE} &= -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \\ \mathcal{L} &= \mathcal{L}_{CE} + \lambda \sum_{\theta \in \Theta} \|\theta\|_2^2 \end{aligned} \quad (14)$$

Furthermore, to mitigate the effects of class imbalance, we employ a weighted cross-entropy loss function during training, where the weight for each class is set to be inversely proportional to its frequency in the training data. We selected class weighting as our sole strategy, as it directly adjusts the model’s focus without altering the original data distribution, a risk associated with resampling methods. We use the Adam optimizer with learning rate scheduling and an early stopping strategy to prevent overfitting.

4. Experiments

4.1. Experimental setup

To validate the effectiveness of our proposed framework, we conduct extensive experiments designed to answer the following research questions:

1. How does our framework perform against state-of-the-art rumor detection methods?
2. How crucial are the various components of our model, particularly the commonsense knowledge module, to its overall performance?
3. How effective is our framework in the challenging reply-free setting, where it must rely solely on the source post and its propagation dynamics?

Table 1
Statistical information of the PHEME datasets.

Events name	Rumors	Non-rumors	Source posts
Charlie Hebdo	458	1621	2079
Sydney Siege	522	699	1221
Ferguson	284	859	1143
Ottawa Shooting	470	420	890
Germanwings Crash	238	231	469
Total	1972	3830	5802

Table 2
Statistical information of the Twitter15 and Twitter16 datasets.

Details	Rumors	Non-rumors	Source posts
Twitter15	372	370	742
Twitter16	207	205	412

This section details the datasets, evaluation metrics, baseline models, and implementation specifics of our experimental protocol. We also present a comprehensive analysis of the results, including ablation studies and case studies, to thoroughly assess our method's capabilities.

4.2. Datasets

We evaluate our model on three widely-used public benchmark datasets: PHEME, Twitter15, and Twitter16 [29–31].

- **PHEME:** The PHEME dataset contains 5802 source posts with their reactions and metadata of five breaking news events, where source posts are annotated as rumors or non-rumors. Professional journalists from partner news organizations provided ground truth verification. Each source tweet was fact-checked against authoritative sources (government agencies, verified news outlets, official statements). This verification process included a crucial check for temporal consistency, ensuring that claims were evaluated based on the information and context available at the time of posting.
- **Twitter15 and Twitter16:** Twitter15 and Twitter16 datasets contain rumors collected from Twitter. The ground-truth labels (true, false, unverified, non-rumor) were determined based on the verification results from authoritative fact-checking websites such as Snopes.com and Emergent.info, where we utilize only “true”, “false” instances for binary classification.

These datasets' URLs were replaced with “URL” and user mentions with “USER”. Hashtags were retained for analysis. All emojis were converted to their textual codes using the Python emoji library. Parent-child relationships were established based on reply or retweet metadata, strictly enforcing temporal consistency (child timestamp must be later than parent). To ensure meaningful analysis, all propagation trees with fewer than 3 nodes were discarded. A core aspect of our evaluation is the strict adherence to the reply-free setting. Throughout all experiments, our model is exclusively provided with the source post content and its propagation structure; no information from user replies is utilized. Detailed statistics for the datasets are provided in Tables 1 and 2.

4.3. Evaluation metrics

We assess the performance of all methods using three standard classification metrics: Precision, Recall, and F1-score. Precision is critical because a low score (a high number of False Positives) means legitimate news or opinions are incorrectly flagged as rumors, which could lead to undesirable censorship and undermine public trust. Recall is equally vital, as a low score (a high number of False Negatives) signifies that harmful rumors are missed by the detection system, allowing misinformation to propagate freely and cause societal harm. Given this trade-off, the F1-score serves as our primary metric because it provides a balanced

assessment of a model's ability to be both accurate in its positive predictions and comprehensive in identifying all actual rumors. The formulas are defined in Eqs. 15–17.

$$Precision = \frac{TP}{TP + FP} \quad (15)$$

$$Recall = \frac{TP}{TP + FN} \quad (16)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (17)$$

where TP, FP, and FN represent the counts of true positives, false positives, and false negatives respectively. While these metrics are standard, it is important to critically consider their properties. For instance, in datasets with class imbalance, a high F1-score can sometimes be driven by strong performance on the majority class. Although our datasets are relatively balanced (see Tables 1–2), we mitigate this potential issue by employing a weighted cross-entropy loss function during training. Furthermore, these metrics do not capture the full picture of a model's real-world utility, such as its interpretability or computational efficiency, which we evaluate separately in Sections 5.5 and 5.7. Despite these limitations, Precision, Recall, and F1-score remain the most effective and widely-accepted metrics for providing a balanced and comparable assessment of classification performance in the rumor detection literature. For this binary classification task, all reported metrics are calculated based on these standard formulas, equivalent to a macro-averaged score.

4.4. Baselines

We compare our proposed framework against a comprehensive set of baseline and state-of-the-art (SOTA) methods. To ensure a rigorous and fair evaluation of K-FLARE's capabilities in its target scenario, all baseline models were adapted and evaluated under the same strict reply-free constraint. For methods that traditionally rely on user reply content, we modified their feature sets or inputs to exclude any information derived from replies. For instance, features related to the text of replies (e.g., sentiment, topics) were removed, and graph-based models were constructed using only the propagation structure without access to the content of reply nodes. This approach creates a level playing field, allowing us to directly assess the efficacy of each model in compensating for the absence of reply-based signals, which is the core challenge this work addresses.

- **DTC(2011):[11]** A feature-based method that uses hand-crafted statistical features from posts and user interactions to assess credibility.
- **PTK(2017):[32]** A kernel-based approach that captures high-order structural patterns by evaluating similarities between rumor propagation trees.
- **Zubiaga et al.(2017):[33]** A method that uses linear-chain conditional random fields to leverage the sequential dynamics of social context for rumor detection.
- **Abulaish et al.(2019):[34]** A graph-based approach that models user-generated content and utilizes sentimental terms, such as anxiety and doubt, to detect rumors.
- **Ajao et al.(2019):[35]** A sentiment-aware method that uses the ratio of negative to positive emotions within social media posts to detect misinformation.
- **Lathiya et al.(2020):[36]** This approach models the sequential dynamics of posts and leverages user homophily as a key feature for rumor detection.
- **Haque and Abulaish(2022):[37]** A graph-based method that extracts underlying rumor patterns from posts and reactions, enhanced by contextual and semantic features from word embeddings.

Table 3

Hyperparameter settings used for training the K-FLARE framework.

Hyperparameter	Value
Semantic Embedding Dimension	768
Topological Feature Dimension	9
Knowledge Embedding Dimension	768
Temporal Feature Dimension	6
Hidden Dimension (d_{hidden})	256
Number of Attention Heads (h)	4
Random Seeds	42
Optimizer	Adam
Initial Learning Rate	1e-4
Batch Size	20
Max Epochs	100 (with early stopping)
Dropout Rate	0.3
L2 Regularization (λ)	1e-5

Table 4

Comparative performance on the PHEME dataset. Results are reported as mean $\pm$ standard deviation over 5 runs.

Model	Precision	Recall	F1-score
DTC	71.52 $\pm$ 0.85	53.15 $\pm$ 1.10	60.98 $\pm$ 0.95
PTK	69.00 $\pm$ 0.92	65.05 $\pm$ 0.88	66.96 $\pm$ 0.90
Zubiaga et al. (2017)	31.21 $\pm$ 1.25	41.55 $\pm$ 1.30	35.67 $\pm$ 1.28
Abulaish et al. (2019)	38.74 $\pm$ 1.15	39.81 $\pm$ 1.20	39.27 $\pm$ 1.18
Ajao et al. (2019)	78.91 $\pm$ 0.65	80.23 $\pm$ 0.58	79.56 $\pm$ 0.60
Lathiya et al. (2020)	46.12 $\pm$ 0.98	48.33 $\pm$ 1.05	47.20 $\pm$ 1.01
Haque and Abulaish (2022)	85.43 $\pm$ 0.50	81.12 $\pm$ 0.55	83.22 $\pm$ 0.52
Rezaei et al. (2023)	82.67 $\pm$ 0.61	79.54 $\pm$ 0.65	81.07 $\pm$ 0.63
Haque and Abulaish (2023)	83.55 $\pm$ 0.58	81.76 $\pm$ 0.62	82.64 $\pm$ 0.59
CARES (Haque and Abulaish, 2025)	84.15 $\pm$ 0.45	84.21 $\pm$ 0.41	84.12 $\pm$ 0.42
K-FLARE	84.41 $\pm$ 0.38	91.19 $\pm$ 0.35	87.64 $\pm$ 0.36

- **Rezaei et al.(2023):[38]**A feature-centric approach that utilizes a combination of textual, sentiment, semantic, and readability features for detection.
- **Haque and Abulaish(2023):[39]**An approach that combines emotion-related aspects, psycholinguistic features, and word embeddings to detect rumors.
- **CARES (Haque and Abulaish, 2025):[19]**A SOTA method that integrates graph-based contextual learning, commonsense knowledge from ConceptNet, and word embeddings to model rumor propagation.

4.5. Implementation details

Text Preprocessing: All textual content undergoes a preprocessing stage where redundant whitespace is removed. Crucially, punctuation, emojis, stop words, and capitalization are preserved, as they provide important signals regarding user writing style, emphasis, and syntactic structure that are valuable for our analysis.

Model Configuration: For the LLM Content Analysis module, we utilize the deepseek-v1:14b model [40]. Our choice was motivated by its strong performance on reasoning and instruction-following benchmarks, which are critical for the structured content analysis task central to our framework. The inference process was configured with a temperature of 0.8 and a maximum token limit of 1000. The semantic feature extraction was performed using the bert-base-uncased model [41], with a maximum sequence length set to 128. For the zero-shot classification task, we adopt a standard Natural Language Inference (NLI) approach using the bart-large-mnli model from Hugging Face [42]. This model is widely recognized for its strong performance on NLI tasks and serves as a robust baseline for zero-shot text classification.

Hyperparameter Settings: The neural fusion module was trained end-to-end using the Adam optimizer. Key hyperparameters for our model are detailed in Table 3.

Table 5

Comparative performance on the Twitter15 dataset. Results are reported as mean $\pm$ standard deviation over 5 runs.

Model	Precision	Recall	F1-score
DTC	62.15 $\pm$ 0.71	72.81 $\pm$ 0.55	67.05 $\pm$ 0.62
PTK	74.52 $\pm$ 0.48	69.26 $\pm$ 0.63	71.80 $\pm$ 0.51
Zubiaga et al. (2017)	34.53 $\pm$ 0.95	46.21 $\pm$ 1.12	39.55 $\pm$ 1.05
Abulaish et al. (2019)	46.88 $\pm$ 0.88	36.14 $\pm$ 1.25	40.81 $\pm$ 0.98
Ajao et al. (2019)	75.31 $\pm$ 0.52	64.55 $\pm$ 0.68	69.51 $\pm$ 0.59
Lathiya et al. (2020)	47.16 $\pm$ 0.75	49.88 $\pm$ 0.81	48.48 $\pm$ 0.77
Haque and Abulaish (2022)	79.54 $\pm$ 0.45	75.21 $\pm$ 0.49	77.31 $\pm$ 0.46
Rezaei et al. (2023)	76.88 $\pm$ 0.51	73.01 $\pm$ 0.58	74.89 $\pm$ 0.53
Haque and Abulaish (2023)	78.11 $\pm$ 0.49	73.23 $\pm$ 0.55	75.59 $\pm$ 0.51
CARES (Haque and Abulaish, 2025)	80.15 $\pm$ 0.41	82.43 $\pm$ 0.38	81.27 $\pm$ 0.35
K-FLARE	91.95 $\pm$ 0.33	92.06 $\pm$ 0.29	92.01 $\pm$ 0.28

Table 6

Comparative performance on the Twitter16 dataset. Results are reported as mean $\pm$ standard deviation over 5 runs.

Model	Precision	Recall	F1-score
DTC	65.23 $\pm$ 0.68	58.71 $\pm$ 0.75	61.80 $\pm$ 0.70
PTK	65.45 $\pm$ 0.72	69.04 $\pm$ 0.65	67.19 $\pm$ 0.68
Zubiaga et al. (2017)	41.89 $\pm$ 1.10	48.11 $\pm$ 1.15	44.77 $\pm$ 1.12
Abulaish et al. (2019)	41.56 $\pm$ 1.21	49.03 $\pm$ 1.18	44.98 $\pm$ 1.19
Ajao et al. (2019)	73.01 $\pm$ 0.55	62.15 $\pm$ 0.62	67.14 $\pm$ 0.58
Lathiya et al. (2020)	48.21 $\pm$ 0.81	49.32 $\pm$ 0.85	48.76 $\pm$ 0.83
Haque and Abulaish (2022)	69.15 $\pm$ 0.60	70.05 $\pm$ 0.58	69.60 $\pm$ 0.59
Rezaei et al. (2023)	70.11 $\pm$ 0.58	66.21 $\pm$ 0.63	68.10 $\pm$ 0.60
Haque and Abulaish (2023)	72.80 $\pm$ 0.56	69.43 $\pm$ 0.59	71.07 $\pm$ 0.57
CARES (Haque and Abulaish, 2025)	82.41 $\pm$ 0.38	83.77 $\pm$ 0.35	83.08 $\pm$ 0.36
K-FLARE	92.77 $\pm$ 0.31	92.39 $\pm$ 0.34	92.58 $\pm$ 0.32

Training Protocol: We employ a hold-out validation strategy, splitting each dataset into training and testing sets with an 80:20 ratio. This ratio was chosen for consistency with the baseline models, which ensures a fair comparison of results. For robust model training and effective hyperparameter tuning, 5-fold cross-validation was performed on the 80% training set. The final model performance is reported on the independent test set. All methods were implemented in Python.

5. Results and analysis

5.1. Overall performance

To address RQ1, we present the comparative performance of K-FLARE against all baseline and SOTA methods across the three benchmark datasets. The detailed results for Precision, Recall, and F1-score are summarized in Table 4 for the PHEME dataset, Table 5 for the Twitter15 dataset, and Table 6 for the Twitter16 dataset. All results in Tables 4, 5, and 6 are reported as mean $\pm$ standard deviation over 5 independent runs.

The results demonstrate that K-FLARE framework shows a clear and consistent performance superiority across all three benchmark datasets. On the Twitter15 and Twitter16 datasets, K-FLARE significantly outperforms all baselines, achieving mean F1-scores of 92.01% and 92.58%, respectively. This highlights the effectiveness of our multimodal fusion strategy in general-context, low-interaction scenarios. Notably, on the challenging PHEME dataset, after ensuring a level playing field, K-FLARE also surpasses all competitors with a robust F1-score of 87.64%. These results validate that our framework, specifically designed for reply-free conditions, is not only effective but superior to adapted state-of-the-art methods when user reply content is unavailable.

5.2. Discussion

Our analysis shows that K-FLARE’s performance depends on the dataset’s characteristics, which aligns with our framework’s design.

Dataset Characteristics: K-FLARE’s superior performance on Twitter15 and Twitter16 stems from its ability to capitalize on the intrinsic signals within the source post itself. In these low-interaction scenarios, where reply-based cues are absent, K-FLARE’s deep semantic analysis (from BERT) and external knowledge verification (from ConceptNet) provide a decisive advantage. In contrast, traditional methods like DTC, which rely heavily on interaction-based statistical features, are naturally disadvantaged. On the PHEME dataset, which covers fast-moving breaking news, the performance gap narrows. This dataset is known to have richer reply content with strong emotional and skeptical cues (e.g., “unconfirmed”). Advanced reply-dependent models like CARES are designed to excel in such data-rich environments. While our adapted, reply-free version of CARES performs well, K-FLARE remains highly competitive and even surpasses it. This demonstrates K-FLARE’s robustness, it achieves state-of-the-art performance without the signals that its competitors were designed to rely on.

Robustness and Scenario Adaptability: This result highlights a key advantage of our approach. In real-world scenarios, especially during the first few hours of a rumor’s lifecycle, reply data is sparse or non-existent. In this critical window, the performance of reply-dependent models would degrade significantly. K-FLARE, by design, is built for this exact scenario, making it a more practical and reliable tool for early intervention systems. Furthermore, these results explicitly validate our architectural design as outlined in Section 1.3. The superior performance, especially on Twitter15 and Twitter16, is directly attributable to our Dual-Stream Hybrid Reasoning Framework. The data-driven inductive stream and the knowledge-driven deductive stream provide complementary signals that are critical in the absence of replies. This synergy is confirmed by the module-level ablation study, which shows that removing either the Hybrid Model for Language Reasoning module or the Neural Fusion module causes a catastrophic performance drop. This validates the necessity of both streams and the effectiveness of our Evidence-Balancing Mechanism in integrating their evidence for a robust final prediction.

5.3. Ablation study

To quantify the contribution of each component to the overall performance (RQ2 and RQ3), we conducted a comprehensive ablation study. We systematically remove key modules and individual features to analyze their impact on precision and F1-score.

Impact of Core Modules: To evaluate the contribution of the main architectural pillars of K-FLARE, we conducted a module-level ablation study. We systematically deactivated three core components—the Neural Fusion Module, the LLM Content Analysis Module, and the NLI Zero-shot Classification Module—and measured the impact on both Precision and F1-scores across the three datasets. The results are illustrated in Figs. 5 and 6.

- Removing the Neural Fusion Module, which integrates the four semantic, knowledge-based, topological and temporal features, causes the most significant and consistent drop in performance. The F1-score plummets by over 20 points on all datasets (to 64.83 % on Twitter15, 57.14 % on Twitter16, and 71.96 % on PHEME). Similarly, Precision scores see a catastrophic decline, dropping by approximately 25 %-30 %. This confirms that the feature fusion backbone is the single most critical component for compensating for the absence of reply content.
- Removing the LLM Content Analysis Module results in a consistent and significant performance degradation, demonstrating its role as a crucial refining component. This leads to an F1-score drop of approximately 3–4 points across all datasets (e.g., from 92.58 % to 88.48 %

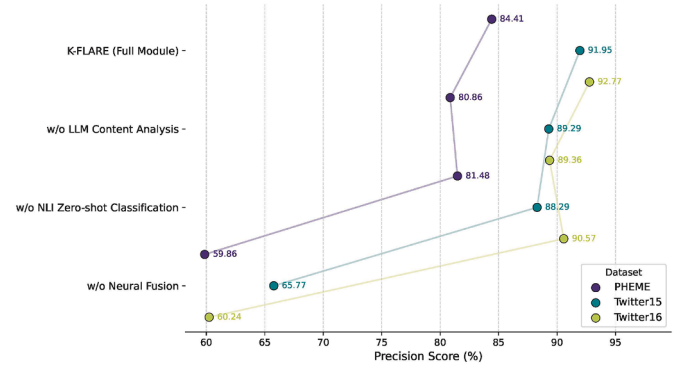


Fig. 5. Ablation study results for Precision on the three datasets. “K-FLARE (Full Module)” is our complete proposed model. “w/o LLM Content Analysis” removes the structured analysis from the Hybrid Model for Language Reasoning Module, “w/o NLI Zero-shot Classification” removes Zero-shot Classification Module and “w/o Neural Fusion” removes all features from the Neural Fusion Module (semantic, knowledge, topological and temporal).

on Twitter16). The impact on Precision is also notable, particularly on the PHEME dataset, where it drops from 84.41 % to 80.86 %. This indicates that the deep semantic reasoning and feature identification provided by the large language model are vital for achieving state-of-the-art accuracy.

- Removing the NLI Zero-shot Classification Module is also detrimental to the model’s performance on all datasets. The effect on the F1-score is significant, with drops to 88.42 % on Twitter15 and 88.21 % on Twitter16. On the PHEME dataset, its removal leads to a smaller but still clear decrease in performance from 87.64 % to 85.90 %. This indicates that the NLI’s probabilistic signal provides a valuable, complementary signal that enhances the model’s predictive power in all tested scenarios.

These results validate K-FLARE design philosophy of leveraging a multi-component architecture. The Neural Fusion Module serves as the essential foundation, while both the LLM Content Analysis and NLI Zero-shot Classification contribute distinct and crucial signals, demonstrating a successful synergy that enhances the framework’s overall effectiveness.

Impact of Individual Features: To further dissect the Neural Fusion Module and understand the contribution of each information source, we conducted a fine-grained ablation study by deactivating one feature type at a time. The results, averaged over 5 runs for robustness, are presented in Table 7. The analysis reveals several key insights into the framework’s mechanics:

- Removing *semantic features* causes the most significant performance degradation, with the F1-score dropping from 92.01 % to 84.35 %. This confirms that the semantic content of the source post is the foundational pillar for classification in the challenging reply-free setting.
- Deactivating *knowledge* and *topological features* also leads to a notable decline in performance. This demonstrates their critical role in providing essential external context and structural cues that the model leverages to make more accurate predictions.
- The removal of *temporal features* results in the smallest impact, suggesting that for these datasets, the static semantic and topological features are more discriminative than the dynamics of propagation.

This analysis reveals a clear synergy between the modules. For instance, while the semantic features form the backbone, the knowledge and topological features provide essential context that elevates performance. This demonstrates that no single modality is sufficient in the challenging reply-free setting and validates our design of fusing complementary information sources.

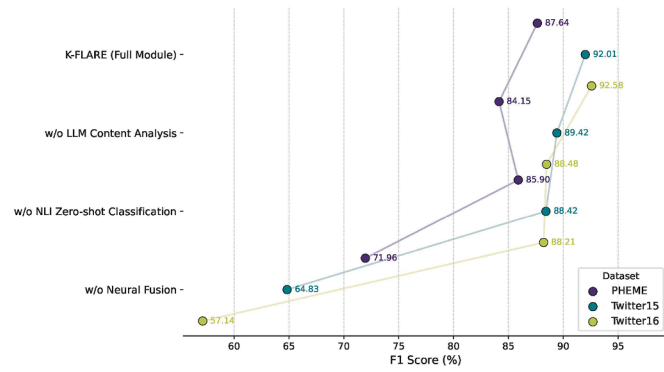


Fig. 6. Ablation study results for F1-score on the three datasets. Each variant is compared to the full K-FLARE module to show the contribution of the LLM content analysis, NLI Zero-shot Classification and the Hybrid Model for Language Reasoning Module.

Table 7

Impact of removing individual feature types on the Twitter15 dataset. Results are reported as the mean $\pm$ standard deviation over 5 runs.

Configuration	Precision	F1-score
Full Feature Model	91.95 $\pm$ 0.33	92.01 $\pm$ 0.28
w/o Semantic Feature	84.56 $\pm$ 0.51	84.35 $\pm$ 0.45
w/o Knowledge Feature	89.26 $\pm$ 0.45	89.47 $\pm$ 0.39
w/o Topological Feature	88.59 $\pm$ 0.48	88.74 $\pm$ 0.41
w/o Temporal Feature	89.26 $\pm$ 0.35	89.33 $\pm$ 0.30

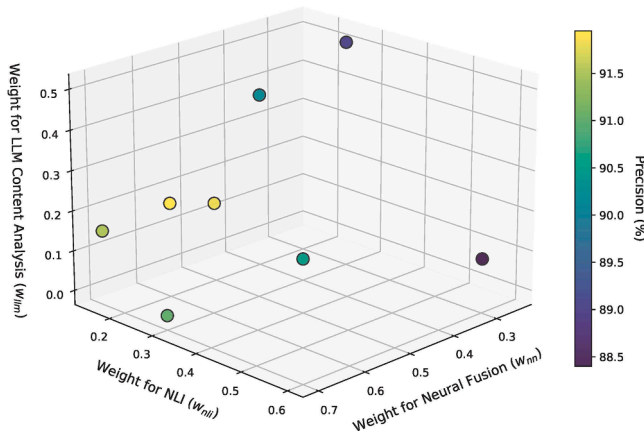


Fig. 7. Sensitivity Analysis of Decision Module Weights on Precision Score (Twitter15 Dataset).

5.4. Parameter sensitivity analysis

We analyzed the model's sensitivity to key hyperparameters to understand their impact on performance.

Decision Module Weights: As shown in Figs. 7 and 8, the weights (w_1, w_2, w_3) in the final decision module significantly affect performance. The results indicate that the Neural Fusion Model output (w_3) consistently provides the strongest contribution. While the NLI's zero-shot prediction (w_1) has lower standalone accuracy, it serves as an effective supplement when assigned a moderate weight. An excessively high weight for the LLM content analysis confidence (w_2) can sometimes dilute the final prediction, suggesting that a balanced configuration yields the best results.

Architectural Hyperparameters: We investigated the impact of the number of heads (h) in the multi-head attention mechanism and the hidden layer dimension (d_{hidden}). We found that $h=4$ provides an optimal

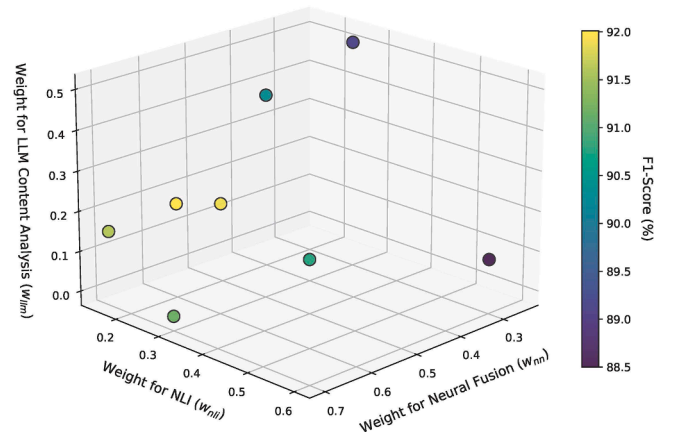


Fig. 8. Sensitivity Analysis of Decision Module Weights on F1-score (Twitter15 Dataset).

balance between representational capacity and model complexity. For the hidden dimension, $d_{hidden} = 256$ achieved the best performance, as smaller dimensions limited the model's capacity and larger dimensions increased the risk of overfitting.

5.5. Case study: LLM content analysis

Fig. 9 presents a case study that highlights the LLM Content Analysis Module's interpretability. For the rumor post, a sensational claim about a "mysterious creature"—the module correctly classifies it with high confidence. Its justification is transparent, listing classic rumor indicators like "The use of emotional language" and "Lack of verifiable sources". This demonstrates the model's ability to ground its decisions in well-established misinformation cues. The non-rumor case, concerning a breaking news event, reveals a more sophisticated reasoning process. The input is "breaking news: soldier shot at national war memorial in ottawa". The LLM correctly identifies that the message contains features often associated with rumors, such as a "Sense of urgency or time pressure" (from "breaking news") and a "Lack of verifiable sources" within the post itself. However, the LLM does not blindly classify it as a rumor based on these superficial cues. As its explanation reveals, it leverages its internal world knowledge to recognize that the content is "consistent with known events (e.g., the 2014 Ottawa shooting)". By verifying the claim's factual basis, the model correctly overrides the heuristic indicators and assigns a low probability of it being a rumor. This case study demonstrates the model's critical capability to distinguish urgent, factual reporting from fabricated rumors during early-stage event analysis. Ultimately, these cases show the LLM content analysis module provides more than just a classification. It offers a transparent reasoning process, using its background knowledge to distinguish fact from rumor even when superficial cues are ambiguous. This enhances not only the accuracy but also the trustworthiness of the entire framework. This transparent reasoning process is a key advantage for practical applications, as it allows human moderators to understand the basis for the model's prediction, fostering trust and enabling more informed final decisions. However, it is important to note that prompt-based explanations, while plausible, do not guarantee correctness and the internal workings of the LLM remain partially opaque.

5.6. Error analysis and limitations

To better understand the boundaries of K-FLARE's capabilities, we conducted a qualitative analysis of its misclassifications. This error analysis reveals specific scenarios where the model struggles, providing valuable insights for future improvements.

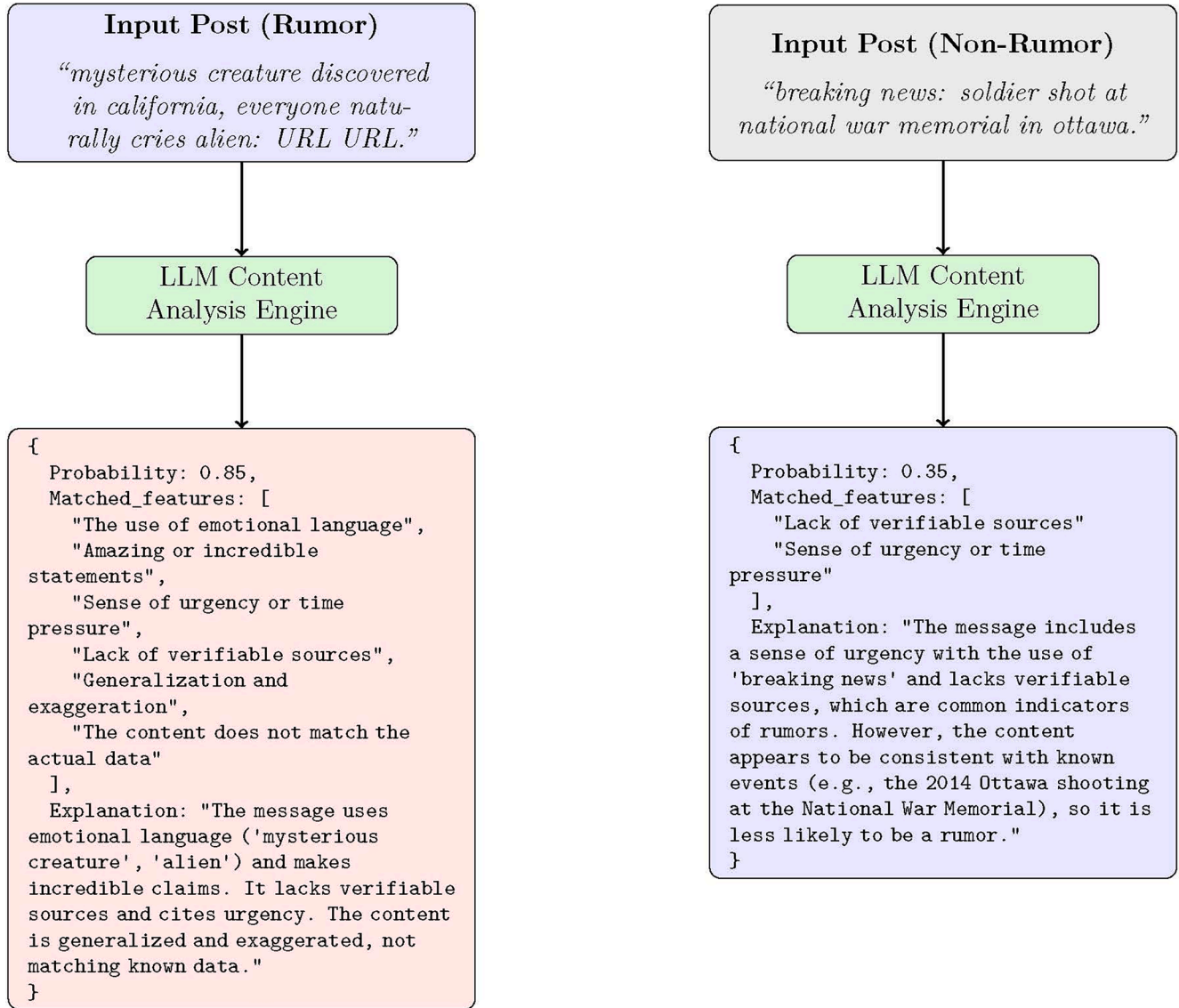


Fig. 9. Case study of the LLM Content Analysis Module on a sample rumor and a non-rumor post. The module takes a raw post as input and generates a structured JSON analysis.

- **False Positive Example:** “Steve Jobs was adopted. His biological father was Abdulfattah Jandali, a Syrian Muslim”. K-FLARE correctly identified that the core statements in the post are factually true. Steve Jobs was indeed adopted, and Abdulfattah Jandali was his biological father. Because the semantic content and its corresponding commonsense knowledge graph did not reveal a direct contradiction, K-FLARE assigned a very low probability of it being a rumor. However, K-FLARE failed to recognize the malicious framing. The inclusion of the seemingly irrelevant detail “a Syrian Muslim” is a common technique in misinformation designed to stoke xenophobia and create a misleading association. The post’s deceptiveness lies not in its facts, but in its intent. This reveals a critical limitation: K-FLARE excels at verifying facts but struggles to infer authorial intent or recognize when true information is being used for malicious narrative-building. It currently lacks the nuanced understanding to detect this form of “truthful misinformation”.
- **False Negative Example:** “pope francis again appears to suggest progress, but just makes false claims, as his religious belief requires of him URL”. K-FLARE correctly identified several stylistic red flags that are highly correlated with rumors: strong emotional and accusatory language (“false claims”), broad generalizations about a

group’s motives (“as his religious belief requires of him”), and a lack of immediately verifiable sources. These features prompted the LLM Content Analysis module to assign a high rumor probability. However, the post is fundamentally an expression of strong, critical opinion, not a verifiable factual claim. The “false claims” mentioned are part of a subjective critique rather than a specific, debunkable piece of misinformation. This case highlights the K-FLARE’s difficulty in distinguishing between fact-based misinformation and highly subjective, polemical commentary. K-FLARE’s pattern recognition, while effective against many rumors, can misfire when faced with opinion pieces that adopt an aggressive or sensationalist tone.

These findings demonstrate that K-FLARE’s primary limitations are not in its core feature extraction, but in higher-level semantic interpretation. Specifically, the framework is challenged by: (1) **Intent vs. Content:** It can validate the factual content of a message but may miss malicious intent when true facts are used to mislead. (2) **Opinion vs. Fact:** It can confuse the stylistic attributes of strongly-worded opinions with the features of fact-based rumors. These insights directly inform our future work. Addressing these limitations highlights challenges not only in model accuracy but also in interpretability; the model must not

Table 8
Average inference time per post on the Twitter15 test set. Time is measured in millisecond.

Model	Average Inference Time (ms/post)
DTC	2.74
BiGCN	7.23
K-FLARE	22.58

only be correct, but its reasoning (e.g., distinguishing intent from literal meaning) must also be understandable and verifiable. Future work will require moving beyond fact-verification and stylistic analysis towards the development of quantitative metrics for author intent detection and the robust classification of a statement as either a verifiable claim or a subjective opinion, coupled with methods to make these nuanced distinctions transparent.

5.7. Computational efficiency

To assess the practical deployment feasibility of K-FLARE, we evaluate its computational efficiency by measuring the average inference time per post. The experiments were conducted on a workstation equipped with a single NVIDIA GeForce RTX 4060 Ti GPU with 16GB of VRAM. We compare the performance of full K-FLARE model against a lightweight traditional baseline (DTC) and a representative BiGCN method (Bian et al. [8]) that does not use an LLM.

As shown in Table 8, the analysis reveals that K-FLARE is remarkably efficient for a framework that incorporates a large language model. With an average inference time of just 22.58 ms per post, it demonstrates a high degree of computational feasibility. While its processing time is numerically higher than the lightweight DTC (2.74 ms) and the GNN-based BiGCN (7.23 ms), the absolute latency is exceptionally low. The primary computational overhead is attributed to the LLM Analysis Module. However, our results indicate that this cost is well-managed, allowing the framework to leverage the power of an LLM without a prohibitive sacrifice in speed. This result highlights a crucial advantage of our approach: K-FLARE achieves its state-of-the-art accuracy and nuanced reasoning capabilities while maintaining high efficiency. An inference time well under 25 ms per post confirms that K-FLARE is highly suitable for real-world, high-throughput deployment on modern hardware, making it not only a powerful but also a practical tool for early rumor intervention.

6. Conclusion and future work

This paper introduced K-FLARE, a novel framework that successfully tackles the critical challenge of reply-free rumor detection. We proposed a new conceptual model, the dual-stream hybrid reasoning architecture, which synergizes a data-driven inductive stream with a knowledge-driven deductive stream. A principled evidence-balancing mechanism integrates these streams, effectively compensating for the information deficit caused by absent user replies. The framework's superiority is demonstrated through extensive experiments on three benchmark datasets, where it achieves state-of-the-art performance, validated by ablation studies and statistical significance tests. Ultimately, K-FLARE provides a robust and interpretable solution for early rumor intervention, advancing the theoretical approach to detection in data-sparse scenarios.

While promising, this work has limitations that point toward important directions for future research. The model was trained exclusively on English Twitter data; therefore, a key priority is to improve its generalization ability across different social media platforms, languages, and domains such as politics and health. We acknowledge that the analyses of interpretability and ethics in this study are qualitative. A critical di-

rection for future work is to develop a quantitative framework for these aspects, addressing the reviewer's call for a formal linkage to decision outcomes. Future work will also focus on enhancing performance in extremely sparse scenarios, such as newly posted content with no propagation data, by exploring advanced prompt engineering and zero-shot learning techniques. Furthermore, we acknowledge the significant ethical challenges inherent in automated rumor detection. The risks of misclassification are non-trivial; falsely flagging legitimate content could lead to censorship, while failing to detect harmful rumors allows misinformation to spread. Specific concerns for K-FLARE include potential biases inherited from the LLM or training data, which could disproportionately affect certain topics or user groups, impacting fairness. The potential for misuse in censorship or through adversarial attacks also requires careful consideration. Therefore, future research must prioritize improving the model's robustness against manipulation and enhancing its fairness across diverse contexts. Deepening interpretability, potentially by generating faithful natural language explanations for "why" a post was flagged, is crucial for accountability. Critically, establishing frameworks for responsible deployment that "always" include human oversight and clear mechanisms for appeal is not merely a suggestion for future work, but a prerequisite for any real-world application in this sensitive domain.

CRediT authorship contribution statement

Jiawei Zhao: Software, Resources, Project administration, Methodology, Investigation, Formal analysis; **Jiangnan Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation; **Yaqin Wu:** Investigation, Conceptualization; **Xiaofeng Cao:** Writing – review & editing, Supervision, Software, Methodology, Investigation.

Data availability

The code and processed data used in this study will be made publicly available on GitHub upon publication of this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. LLM Structured Prompt Template

Below is the structured prompt provided to the LLM for the content analysis task.

Please analyze the following social media messages to determine if they may be rumors and provide a probability value between 0–1.

Please consider the following features:

1. The use of emotional language
2. Amazing or unbelievable statements
3. Sense of urgency or time pressure
4. Lack of verifiable sources
5. Summarize and exaggerate
6. Content inconsistent with the state of knowledge

Return in JSON format, including the following fields: Probability: A value between 0–1, indicating the probability of a rumor. The higher

the probability value, the more likely it is to be a rumor, and vice versa, it is true information

Matched_features: List of matched rumor features

Explanation: A brief explanation

Message content: original post content

Appendix B. Formalization of Evidence-Balancing Mechanism

As introduced in Section 3.5, the weights $\mathbf{w} = (w_{nn}, w_{nli}, w_{llm})$ for the Evidence-Balancing Mechanism (Eq. 12) are not set manually but are learned as optimized hyperparameters. This learning process is formalized as an optimization problem. The objective is to find the optimal set of weights $\mathbf{w}^*$ that minimizes the final cross-entropy loss on the validation folds derived from the training set. This objective function can be formally stated as in Eq. B.1.

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \mathcal{L}_{CE}(p_{final}(\mathbf{w}), y) \quad (\text{B.1})$$

where $\mathbf{w}$ is the vector of weights $(w_{nn}, w_{nli}, w_{llm})$ being optimized. $p_{final}(\mathbf{w})$ is the final predicted probability from Eq. 12. y represents the ground-truth labels in the validation set. This optimization is performed using 5-fold cross-validation on the training data, as described in Section 3.5, to find the $\mathbf{w}$ that provides the most robust and accurate predictions by balancing the evidence from the framework's inductive and deductive streams.

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